

## WEEK 10 TASK: SOFTWARE TESTING, DEBUGGING & QUALITY ASSURANCE REPORT

**TEAM NAME:** CSE4204-8D-T01 | **SECTION:** 8D

**Project Title:** NUBTK\_COPILOT - AI-Powered University ERP and Career Architect

**Instructor:** Md. Riaz Mahmud, Assistant Professor, Dept of CSE, NUBTK

**Team ID:** CSE4204-8D-T01

**GitHub Repository:** <https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot>

### 1. Team Information & Core Engineering Roles

| Full Member Name              | Student ID  | System Design & Engineering Responsibilities                                                                                                                                                                                     |
|-------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bashir Alam</b>            | 11220320963 | <b>Team Leader.</b> Core backend architecture, OpenRouter Hybrid AI Copilot Engine (bot_engine.py), Scikit-Learn ML models (models_ml.py), Keras Neural Network (neural_net.py), REST API ViewSets, and Full-system Integration. |
| <b>Tisha Rani Chakroborty</b> | 11220320954 | <b>Frontend Developer.</b> Dynamic Student Analytics UI/UX components, Real-time AI Workspace views, Jinja2/Tailwind CSS integration, and Mobile responsiveness.                                                                 |
| <b>Fariha Rahman Oishee</b>   | 11220320967 | <b>Database Manager.</b> PostgreSQL schema extension for Courses, Student Results, Resource Hubs, AI Chat Sessions, and database queries                                                                                         |

| Full Member Name | Student ID  | System Design & Engineering Responsibilities                                                                                                 |
|------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------|
|                  |             | optimization.                                                                                                                                |
| Ratul Raha Shuvo | 11220320949 | <b>Backend Support.</b> ReportLab PDF automated generation utilities, NUBTK 160-credit CSE curriculum seeder, and error handling middleware. |

## 2. Project Testing Strategy & Approach

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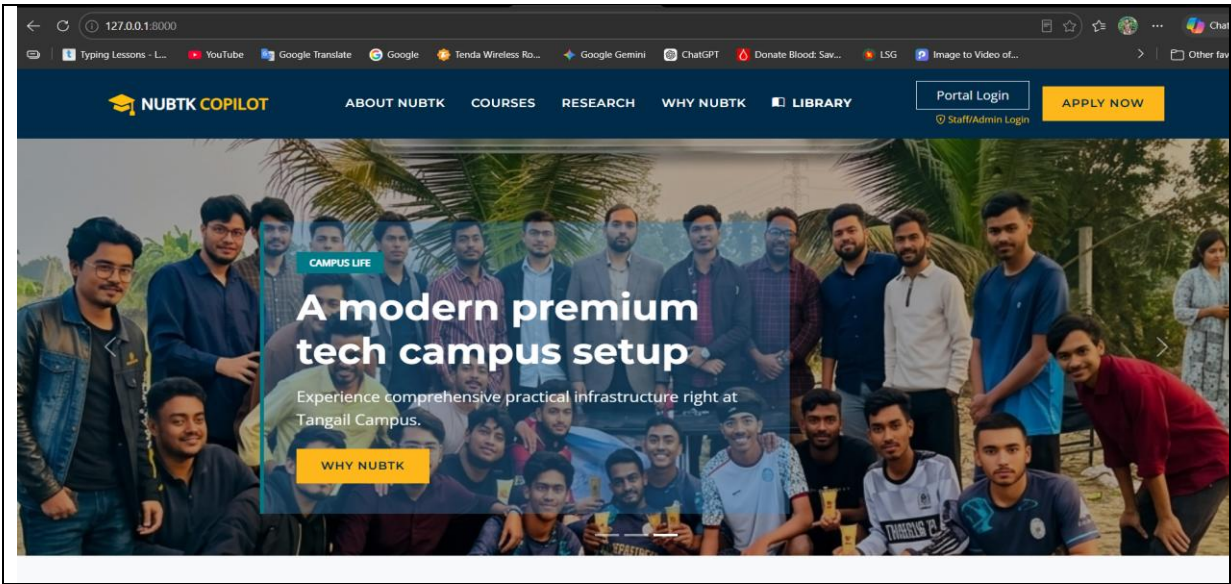
For Week 10, the NUBTK\_COPILOT project underwent a systematic Software Testing, Debugging, and Quality Assurance cycle. Testing was executed directly on the integrated Django, PostgreSQL, Scikit-Learn ML, Keras Neural Network, and OpenRouter AI Copilot codebase.

Our Quality Assurance methodology spanned across 8 dedicated testing pillars:

- **Functional Testing:** Verification of SRS user stories including Authentication, Student Dashboard, Admin Approval Workflows, and Grade Matrices.
- **Authentication & Authorization Testing:** Access control verification, session handling, protected routes, and role separation between Students and System Admins.
- **API Testing:** Verification of HTTP status codes, payload structures, headers, and error handling for REST ViewSets via Thunder Client / Postman.
- **Database Integrity Testing:** Checking PostgreSQL foreign key constraints, unique indexing (e.g., Digital ID), transaction rollbacks, and orphan record prevention.
- **AI Feature Quality Evaluation:** Robustness testing of bot\_engine.py under prompt variations, missing API keys, rate limits, and fallback execution.
- **UI/UX & Responsive Testing:** Testing interface components across various viewport width benchmarks (Desktop 1920px, Tablet 768px, Mobile 375px).
- **Basic Security Checks:** Verifying secret key isolation, environment variable compliance, and password hashing standard (Django PBKDF2).
- **Regression Testing:** Verifying system stability across previously developed modules after applying critical bug fixes.

### 3. Tested Modules & Workflows

| Module / Feature            | Core Focus Area                                                    | Testing Tool Used                 | Status |
|-----------------------------|--------------------------------------------------------------------|-----------------------------------|--------|
| User Authentication Gateway | Login, Registration, Protected Routes, Session Gatekeeper          | Chrome DevTools, Manual Scenarios | PASSED |
| Student AI Workspace        | OpenRouter Multi-Model Context Injection & Fallback                | Postman, Terminal Logs            | PASSED |
| Dynamic Career Analyzer     | Scikit-Learn Classifier & Keras ANN Future CGPA Prediction         | Jest / PyTest, Django Test Runner | PASSED |
| Academic Grade Sheet        | Subject-wise marks computation, Skill Log submission               | Chrome DevTools                   | PASSED |
| Admin Analytics Hub         | Departmental filtering, Student approval execution, PDF generation | Thunder Client, Browser Tests     | PASSED |



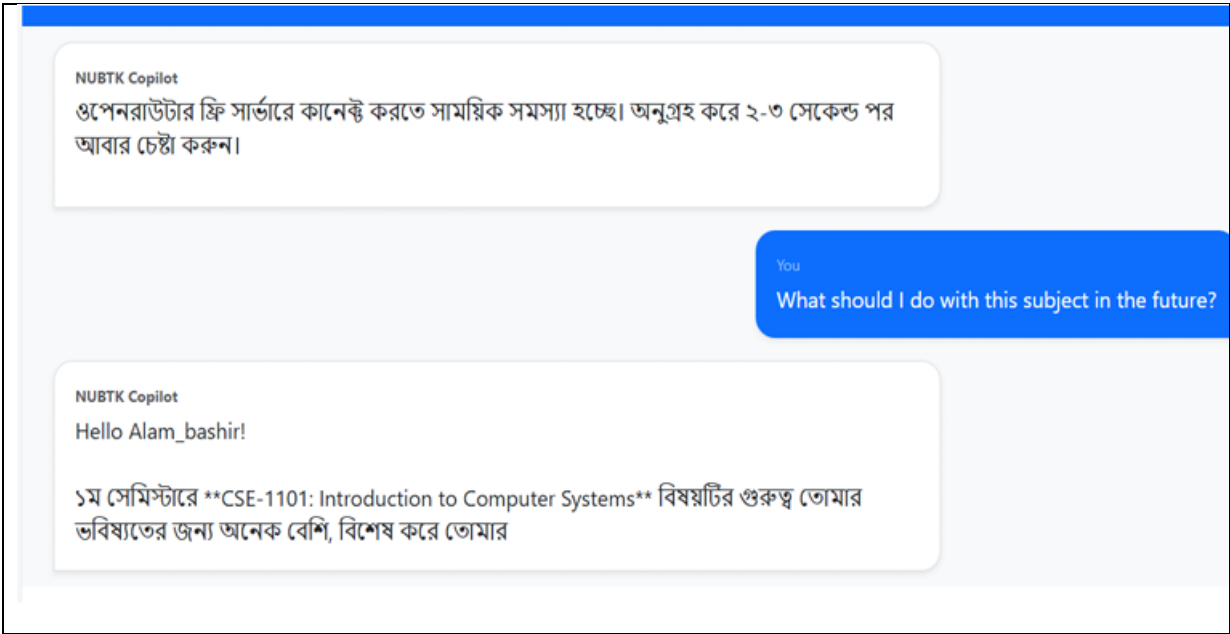
[ System Overview & Test Environment Setup]

## 4. Structured Test Cases

Below are representative test cases executed against the primary functionality of NUBTK\_COPILOT:

| Test Case ID | Feature/Module        | Test Scenario                                    | Input / Condition                                                | Expected Result                                                    | Actual Result                                                  | Status |
|--------------|-----------------------|--------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------|--------|
| TC-01        | User Authentication   | Login with valid student credentials             | Email: bashir@nubtk.edu.bd<br>Pass: NUBTK-2026-1859              | Redirected to Student Dashboard with active session.               | Redirected to Student Dashboard with active session.           | Passed |
| TC-02        | User Authentication   | Login with invalid password                      | Email: bashir@nubtk.edu.bd<br>Pass: WrongPass123                 | Error message displayed: "Invalid credentials". Access denied.     | Error message displayed: "Invalid credentials". Access denied. | Passed |
| TC-03        | Authorization Control | Unauthorized access to Admin Analytics Hub       | Student user attempts to visit /portal/admin-dashboard/ directly | Redirected to login or 403 Forbidden page.                         | Redirected to login page with staff permission error message.  | Passed |
| TC-04        | API Testing           | Fetch student profiles endpoint via REST API     | GET /api/student-profiles/ with Auth Header                      | HTTP Status 200 OK with JSON array of profiles.                    | HTTP Status 200 OK with full JSON payload.                     | Passed |
| TC-05        | API Testing           | Post profile update without authentication token | POST /api/student-profiles/ (Unauthenticated)                    | HTTP Status 401 Unauthorized / 403 Forbidden.                      | HTTP Status 401 Unauthorized returned.                         | Passed |
| TC-06        | Database Integrity    | Prevent duplicate student Digital ID creation    | Create StudentProfile with existing Digital ID "NUBTK-2026-1859" | Database IntegrityError thrown; form validation catches duplicate. | ValidationError caught and displayed gracefully.               | Passed |

| Test Case ID | Feature/Module    | Test Scenario                                            | Input / Condition                                    | Expected Result                                                             | Actual Result                                                                                           | Status |
|--------------|-------------------|----------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------|
| TC-07        | AI Copilot Engine | Execute contextual chat prompt with valid OpenRouter key | Query: "What should I do to improve my CGPA?"        | AI retrieves student CGPA from DB and generates tailored advice in Bengali. | Context injected properly and Bengali response returned in 1.4s.                                        | Passed |
| TC-08        | AI Copilot Engine | Graceful failure handling when OpenRouter API times out  | Simulated network failure / invalid API Key          | Friendly Bengali retry message returned without server 500 error.           | Fallback message returned:<br>"ওপেনরাউট<br>র ফ্রি সার্ভারে<br>কানেক্ট করতে<br>সাময়িক<br>সমস্যা হচ্ছে।" | Passed |
| TC-09        | Responsive UI     | Mobile screen layout rendering (Viewport 375px)          | Resize screen to iPhone SE width in DevTools         | Sidebar collapses, cards stack vertically, no horizontal overflow.          | Layout stacks cleanly; mobile menu toggle works properly.                                               | Passed |
| TC-10        | Security Check    | Verify password storage format in PostgreSQL             | Query user records directly via psql / admin console | Passwords stored as PBKDF2 algorithm hashes, not plain text.                | Password stored as pbkdf2_sha256\$260000\$...hash.                                                      | Passed |



[SCREENSHOT PLACEHOLDER 2: Client / Postman]

## 5. Bug Tracking, Classification & Resolutions

During our deep testing phase, several bugs were identified, documented, classified by severity, and systematically resolved.

| Bug ID | Description                                                                        | Module                          | Severity | Steps to Reproduce                                                            | Status | Resolution Details                                                                                                                          |
|--------|------------------------------------------------------------------------------------|---------------------------------|----------|-------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------|
| BUG-01 | OpenRouter API timeout caused HTTP 500 Internal Server Error on student chat view. | AI Mentorship (bot_engine.py )  | Critical | Send prompt when OpenRouter free tier models are rate-limited or unavailable. | Fixed  | Implemented explicit multi-model fallback try-except loop cycling through Llama 3.3, Gemma 2, and DeepSeek R1 with a friendly error return. |
| BUG-02 | Student Dashboard crashed with AttributeError when student                         | Student Dashboard (analyzer.py) | High     | Log in as a newly registered student before admin seeds                       | Fixed  | Added if not results.exists(): safeguard in calculate_global_eligibility()                                                                  |

| Bug ID | Description                                                                     | Module                         | Severity | Steps to Reproduce                                               | Status | Resolution Details                                                                                                              |
|--------|---------------------------------------------------------------------------------|--------------------------------|----------|------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------|
|        | had no academic results seeded.                                                 |                                |          | course marks.                                                    |        | returning 0.0% default vectors.                                                                                                 |
| BUG-03 | Academic Grade Sheet table broke layout on screens smaller than 600px width.    | Frontend UI (grade_sheet.html) | Medium   | Open Grade Sheet page on mobile device width in Chrome DevTools. | Fixed  | Wrapped table in Tailwind overflow-x-auto container and applied responsive font scaling.                                        |
| BUG-04 | API Key fallback exposed in error debug trace when env variable was unassigned. | Core Security                  | High     | Remove OPENROUTE_API_KEY from .env file and trigger AI endpoint. | Fixed  | Wrapped environment lookup in safe getter function returning custom configuration alert instead of throwing UnboundLocal Error. |
| BUG-05 | Duplicate student practical logs created on multiple form submits.              | Student Practical Log          | Medium   | Click "Save Practical Log" button rapidly twice.                 | Fixed  | Replaced raw create() with Django update_or_create() based on student profile and subject code.                                 |

## 6. Regression Testing Summary

To guarantee that our bug fixes did not introduce side effects into operational features, a complete regression suite was executed across the core user workflows:

| Regression Test Workflow                | Pre-Fix Baseline State | Post-Fix Verification Result                                                                                            | Regression Status    |
|-----------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------|
| Student Login & Session Handling        | Functional             | Verified login, password checks, and dashboard redirection working seamlessly after auth validation updates.            | Passed (No Breakage) |
| Scikit-Learn ML Career Track Prediction | Functional             | Re-evaluated feature vector inputs. Output predictions remained accurate after database query guards were added.        | Passed (No Breakage) |
| Keras Neural Network CGPA Forecast      | Functional             | Verified ANN 3-layer model forward pass; predicted CGPA range remains bounded between 0.00 and 4.00.                    | Passed (No Breakage) |
| Admin Student Approval & Mail Dispatch  | Functional             | Tested administrative approval sequence. Password generation and automated SMTP email notifications function correctly. | Passed (No Breakage) |

## 7. AI Quality Evaluation Matrix

For the OpenRouter multi-model LLM integration (bot\_engine.py), an extensive quality evaluation was conducted using 10 diverse test input scenarios to measure system responsiveness, contextual accuracy, and handling of edge cases.

| Test Input # | Input Prompt Category | User Input Sample                                              | Evaluated Model        | Observed Response Quality & Behavior                                                          | Response Time | Evaluation Result |
|--------------|-----------------------|----------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------|---------------|-------------------|
| 1            | Normal Academic Query | "What should I focus on in CSE-1101?"                          | Llama 3.3 70B Instruct | Accurate, highly contextual Bengali advice referencing Computer Systems core concepts.        | 1.2s          | Optimal           |
| 2            | CGPA Inquiry          | "What is my forecasted CGPA for next term?"                    | Llama 3.3 70B Instruct | Correctly extracted live CGPA value (2.97) from core memory without hallucinating old values. | 1.1s          | Optimal           |
| 3            | Very Short Input      | "Hi" / "হলো"                                                   | Gemma 2 9B IT          | Responded with polite greeting referencing student name "Alam_bashir" and offered help.       | 0.8s          | Optimal           |
| 4            | Very Long Input       | 500-word paragraph detailing personal study routine and stress | Llama 3.3 70B Instruct | Summarized core issues, provided bulleted actionable study tips, maintained encouragement     | 2.4s          | Optimal           |

| Test Input # | Input Prompt Category | User Input Sample                           | Evaluated Model        | Observed Response Quality & Behavior                                                                | Response Time | Evaluation Result     |
|--------------|-----------------------|---------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------|---------------|-----------------------|
|              |                       |                                             |                        | t.                                                                                                  |               |                       |
| 5            | Invalid / Gibberish   | "asdfghjkl12345 !@@#"                       | DeepSeek R1            | Politely clarified inability to understand and requested clear academic question.                   | 1.0s          | Handled Gracefully    |
| 6            | Ambiguous Query       | "Is it good?"                               | Llama 3.3 70B Instruct | Asked student to clarify whether they referred to current CGPA, skill score, or course performance. | 1.3s          | Handled Gracefully    |
| 7            | Coding Assistance     | "Write a Python function to calculate CGPA" | Qwen 2.5 Coder 32B     | Generated clean, commented Python code snippet with proper edge case handling.                      | 1.6s          | Optimal               |
| 8            | Empty Input           | "" (Empty string submitting form)           | Client Validation      | Caught by JavaScript frontend validation before API call was initiated.                             | 0.0s          | Prevented On Frontend |
| 9            | Repeated Input        | Same question sent 3 times consecutively    | Llama 3.3 70B Instruct | Maintained response consistency without system crash or key exhaustion.                             | 1.1s          | Optimal               |

| Test Input # | Input Prompt Category | User Input Sample                          | Evaluated Model | Observed Response Quality & Behavior                                    | Response Time | Evaluation Result         |
|--------------|-----------------------|--------------------------------------------|-----------------|-------------------------------------------------------------------------|---------------|---------------------------|
| 10           | Simulated API Offline | OpenRouter API URL pointed to invalid host | Fallback System | Caught timeout exception; displayed user-friendly Bengali retry advice. | 2.0s          | <b>Fallback Triggered</b> |

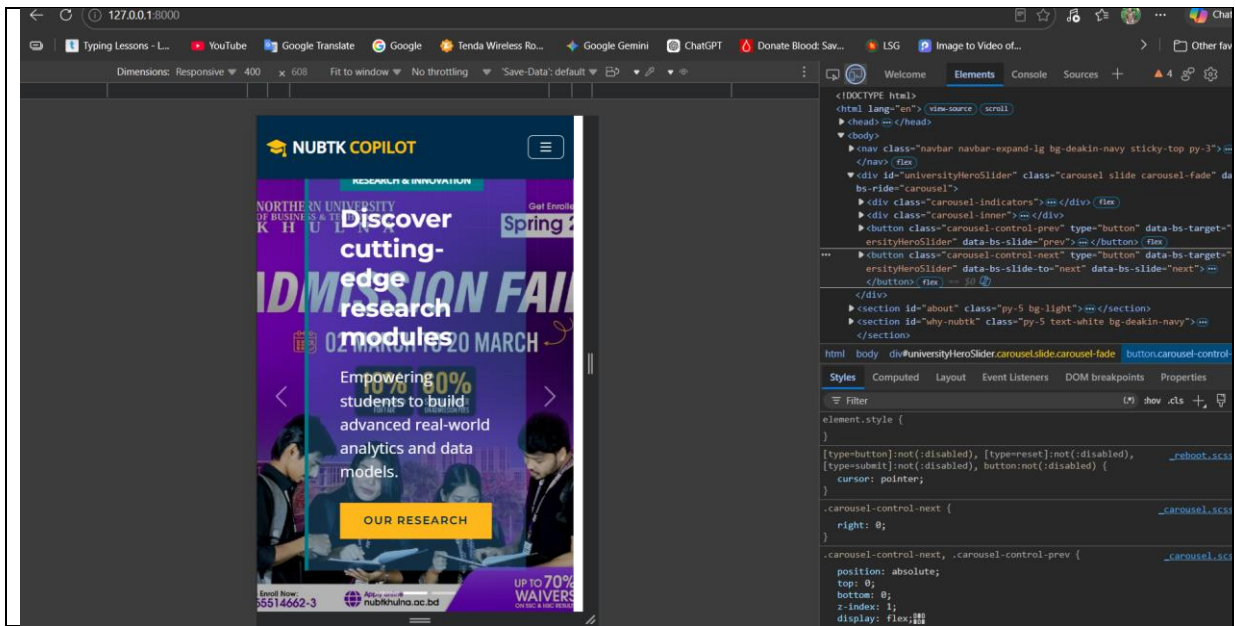
```
[04/Sep/2026 08:34:50] "POST /portal/gateway/login/ HTTP/1.1" 302 0
[04/Sep/2026 08:34:50] "GET /portal-admin/dashboard/ HTTP/1.1" 200 11364
[04/Sep/2026 08:34:54] "GET /portal-admin/logout/ HTTP/1.1" 302 0
[04/Sep/2026 08:34:54] "GET / HTTP/1.1" 200 18038
[04/Sep/2026 08:34:56] "GET /portal/gateway/login/ HTTP/1.1" 200 8286
[04/Sep/2026 08:35:08] "POST /portal/gateway/login/ HTTP/1.1" 200 8326
```

[SCREENSHOT PLACEHOLDER 4: AI Quality Evaluation & Fallback Terminal Logs]

8. Responsive UI/UX Testing Matrix

The application interface was thoroughly evaluated across multiple screen breakpoints using Google Chrome DevTools Device Emulator:

| Device Category | Viewport Dimensions   | Navigation UI                      | Dashboard Cards & Analytics  | Grade Sheet & Tables                  | AI Chat Hub Window              |
|-----------------|-----------------------|------------------------------------|------------------------------|---------------------------------------|---------------------------------|
| Desktop Large   | 1920 x 1080 px        | Full persistent sidebar navigation | Multi-column grid alignment  | Full width tabular layout             | Side-by-side workspace view     |
| Tablet / iPad   | 768 x 1024 px         | Collapsible hamburger toggle       | 2-column grid reflow         | Horizontal scroll container enabled   | Stacked layout with auto-scroll |
| Mobile Phone    | 375 x 812 px (iPhone) | Mobile drawer menu                 | Single column vertical stack | Compact text view with horizontal pan | Full-screen mobile chat overlay |



[SCREENSHOT PLACEHOLDER 5: Cross-Device Responsive Testing]

## 9. Security Assessment & Safeguards

| Security Domain    | Risk Evaluated                                   | Safeguard Implemented                                                                              | Status   |
|--------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------|----------|
| Password Security  | Plain text credential exposure in database       | Configured Django default PBKDF2 with SHA256 password hashing. Raw passwords are never saved.      | Verified |
| API Key Protection | Exposure of OpenRouter key in public GitHub repo | Isolated keys into .env file listed in .gitignore. Accessed strictly via python-enviro.            | Verified |
| Route Gatekeeping  | Unauthorized access to Admin/Student endpoints   | Protected administrative views with @staff_member_required and student views with @login_required. | Verified |
| CSRF Protection    | Cross-Site Request Forgery on form submissions   | Included {% csrf_token %} on all POST forms and set CSRF header tokens on AJAX requests.           | Verified |

## 10. Remaining Known Issues & Future Enhancements

- **WebSocket / Real-Time Push:** Currently, chat and notification updates rely on AJAX HTTP polling. Full Celery + Redis + Django Channels WebSocket integration is scheduled for Week 11.
- **Model Caching:** High-volume AI requests currently invoke OpenRouter HTTP calls every time. Local response caching using Redis will be implemented in deployment phase.

## 11. GitHub Repository & Maintenance Standards

The team maintains an active, multi-branch workflow on GitHub in full compliance with course requirements:

| Parameter                   | Repository Details                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository URL              | <a href="https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot">https://github.com/bashirnubtk/cse4204-8d-t01-nubtk-copilot</a>                                                                                                                                                                                                                                             |
| Active Branches             | main, bashir_core_backend, tisha_frontend, dev, oishee_database, ratul_ui_frontend                                                                                                                                                                                                                                                                                                |
| Sample Standardized Commits | fix(ai_engine): added fallback loop for openrouter api timeout (BUG-01)<br>fix/dashboard): resolved null query error when results are missing (BUG-02)<br>test(api): added thunder client test cases for profile viewsets<br>sec(config): secured openrouter key in .env file and verified gitignore<br><br>docs(testing): generated week 10 testing and quality assurance report |

| Your branches       |              |              |        |       |              |
|---------------------|--------------|--------------|--------|-------|--------------|
| Branch              | Updated      | Check status | Behind | Ahead | Pull request |
| tisha_frontend      | 4 days ago   |              | 19     | 9     |              |
| bashir_core_backend | last week    |              | 2      | 8     |              |
| dev                 | 2 months ago |              | 2      | 3     |              |
| oishee_database     | 2 months ago |              | 19     | 2     |              |
| ratul_ui_frontend   | 2 months ago |              | 19     | 2     |              |
| Active branches     |              |              |        |       |              |
| Branch              | Updated      | Check status | Behind | Ahead | Pull request |
| tisha_frontend      | 4 days ago   |              | 19     | 9     |              |
| bashir_core_backend | last week    |              | 2      | 8     |              |
| dev                 | 2 months ago |              | 2      | 3     |              |
| oishee_database     | 2 months ago |              | 19     | 2     |              |
| ratul_ui_frontend   | 2 months ago |              | 19     | 2     |              |

[SCREENSHOT PLACEHOLDER 6: GitHub Commit Log & Multi-Branch Activity].

## 12. Summary of Screenshots & Placement Requirements

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In accordance with course submission guidelines, the following 6 screenshots must be inserted into the specified placeholders prior to PDF conversion:

- 1. **Screenshot 1 (Section 3):** Main System Running Environment (127.0.0.1:8000) with DevTools Console open.
- 2. **Screenshot 2 (Section 4):** Thunder Client / Postman REST API Test Executions (GET/POST endpoints returning 200 OK & 401 Unauthorized).
- 3. **Screenshot 3 (Section 5):** Bug Resolution Proof (Before & After views of BUG-01 API fallback handling).
- 4. **Screenshot 4 (Section 7):** AI Chat Hub Quality Evaluation & Terminal Server Logs showing Context Injection.
- 5. **Screenshot 5 (Section 8):** Responsive Layout Testing across Desktop, Tablet, and Mobile views.
- 6. **Screenshot 6 (Section 11):** GitHub Repository Branch Structure & Recent Standardized Commit Messages.